Slat control and monitoring system

Description of function

Object code	DMC-MRDX-A-27-81-00-00A-042A-A	**Issue**	003-00 (revised)
Issue date	2026-03-12	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Technical Data Valmont (U8025)
Security	02 — Restricted	**Quality assurance**	First verification (tabtop)
Applicability	MRDX-42 and MRDX-46, all aircraft		

References

CODE	TITLE
DMC-MRDX-A-27-81-00-00A-720A-A	Slat actuation power control unit — Installation
DMC-MRDX-A-27-80-00-00A-040A-A	High lift system — Description

1 General

The slat control and monitoring system moves the leading edge slats between the retracted position and the four extended positions selected on the flap and slat control lever. The system is made of two slat flap control computers (SFCC), one power control unit (PCU) with two hydraulic motors, a torque shaft transmission along the leading edge of each wing, and the wing tip brakes that hold the transmission if an asymmetry is detected.

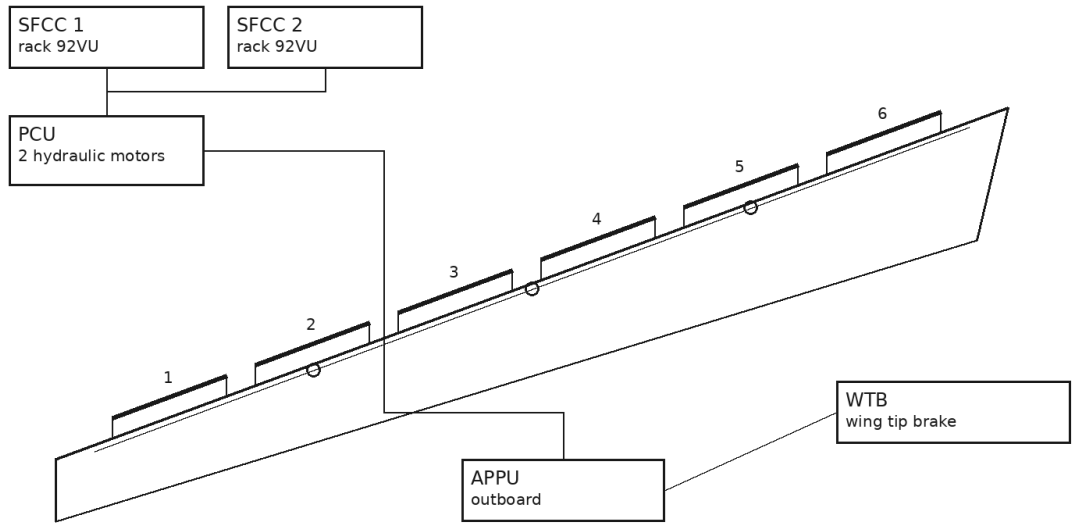
Each SFCC has one slat channel and one flap channel. The two computers work in an active/standby arrangement: the master computer commands the PCU while the standby computer monitors the command and the position feedback. If the master computer fails, control passes to the standby computer without an interruption of the surface movement.

> The position feedback comes from two feedback position pick-off units (FPPU) installed on the PCU output and from one asymmetry position pick-off unit (APPU) installed at the outboard end of each transmission. The APPU signals are compared continuously; a difference of more than 3.6 degrees of transmission rotation between the left and the right wing causes the wing tip brakes to engage **and latches the slat system** until the fault is cleared on the ground.

2 Component location

The PCU is installed in the belly fairing between frame C41 and frame C44, on the left hand side of the centre line. Access is through the fwd belly fairing access panel 196AB. The two SFCC are installed in the avionics compartment, in rack 92VU.

SLAT CONTROL AND MONITORING SYSTEM — COMPONENT LOCATION (LH WING)



1 to 6 slat panels SFCC slat flap control computer PCU power control unit

Slat control and monitoring system — component location — Figure 1

NOTE: The APPU on the right hand wing is accessible only when the slats are fully retracted and the hydraulic systems are depressurized.

3 Operation

3.1 Normal operation

When the flight crew moves the control lever, the two SFCC read the lever position from two independent position sensors. The master computer calculates the commanded slat position, compares it with the FPPU feedback and energizes the PCU valve block until the difference is inside the tolerance band given in Table 1.

Table 1 Slat position tolerance band

Lever position	Slat angle (deg)	Tolerance (deg)	Placard speed (kt)
0	0.0	±0.5	—
1	14.5	±0.8	263
2	19.0	±0.8	222
3	23.0	±1.0	196
FULL	27.0	±1.0	180

3.2 Alternate operation

APPLICABLE TO: Aircraft post-modification 27-8104 embodied

After modification 27-8104, a slat runaway that is detected during the extension sequence no longer causes both hydraulic motors to be isolated. The affected motor is isolated and the remaining motor drives the transmission at half rate. The flight crew gets the `SLATS SLOW` memo on the engine and warning display.

CAUTION

Do not dispatch the aircraft with one slat motor isolated unless the dispatch conditions of the minimum equipment list are satisfied. The landing distance penalty is not applied automatically by the flight management system.

4 Interfaces

- Hydraulic: green and yellow systems, through the PCU valve block.
- Electrical: 28 V DC essential bus for SFCC 1 and 28 V DC bus 2 for SFCC 2.
- Avionics: two AFDX ports per computer to the flight control and guidance function.

Slat actuation power control unit

Installation

Object code	DMC-MRDX-A-27-81-00-00A-720A-A	**Issue**	002-00 (revised)
Issue date	2026-05-04	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Technical Data Valmont (U8025)
Security	01 — Unclassified	**Quality assurance**	Second verification (tabtop)
Applicability	MRDX-42 and MRDX-46, all aircraft		

1. Reason for the job

This data module gives the procedure to install the slat actuation power control unit (PCU) in the belly fairing. Do the procedure after the removal procedure (Ref. Slat actuation power control unit — Removal), or after a PCU has been replaced under modification 27-8117.

2. Job set-up information

A. Referenced information

TYPE	DESIGNATION	IDENTIFICATION No.
DM	The aircraft is safe for maintenance and the landing gear ground locks are installed.	DMC-MRDX-A-00-20-00-00A-941A-A
DM	The green and the yellow hydraulic systems are depressurized.	DMC-MRDX-A-29-00-00-00A-861A-A
GEN	The fwd belly fairing access panel 196AB is open.	—

B. Fixtures, tools, test and support equipment

QTY	DESIGNATION	IDENTIFICATION No.
1	Sling, PCU handling	98D27814500000 (CAGE U8025)
1	Torque wrench, 10 to 70 Nm	TW-70-DR (CAGE 96906)
1	Platform, adjustable, 2 m to 4 m	98D00000100000 (CAGE U8025)

C. Consumable materials

QTY	DESIGNATION	IDENTIFICATION No.
2 l	Hydraulic fluid, phosphate ester, type IV Material No. 02-004	NSA307110 (CAGE 81349)
1 m	Lockwire, corrosion resistant steel, 0.8 mm Material No. 19-003	NSA935403-08 (CAGE 81349)

D. Expendable parts

QTY	DESIGNATION	IDENTIFICATION No.
2	Packing, PCU mounting flange	NSA8102-118 (CAGE U8025)

E. Safety conditions

WARNING

Make sure that the green and the yellow hydraulic systems are depressurized before you disconnect or connect a hydraulic line. Hydraulic fluid at 206 bar (2988 psi) can cause injury to persons and damage to equipment.

CAUTION

Do not let the PCU hang on the hydraulic lines. Damage to the pipe couplings and to the manifold face can occur.

3. Procedure

1. Prepare for the installation

A. Make sure that the mounting flange of the PCU support structure is clean and that no damage is found on the sealing face.

B. Install a new packing NSA8102-118 on each of the two hydraulic manifold ports.

NOTE: Discard the packings that were removed with the unit. Packings are not re-used.

C. Attach the handling sling to the two lifting points on the PCU housing.

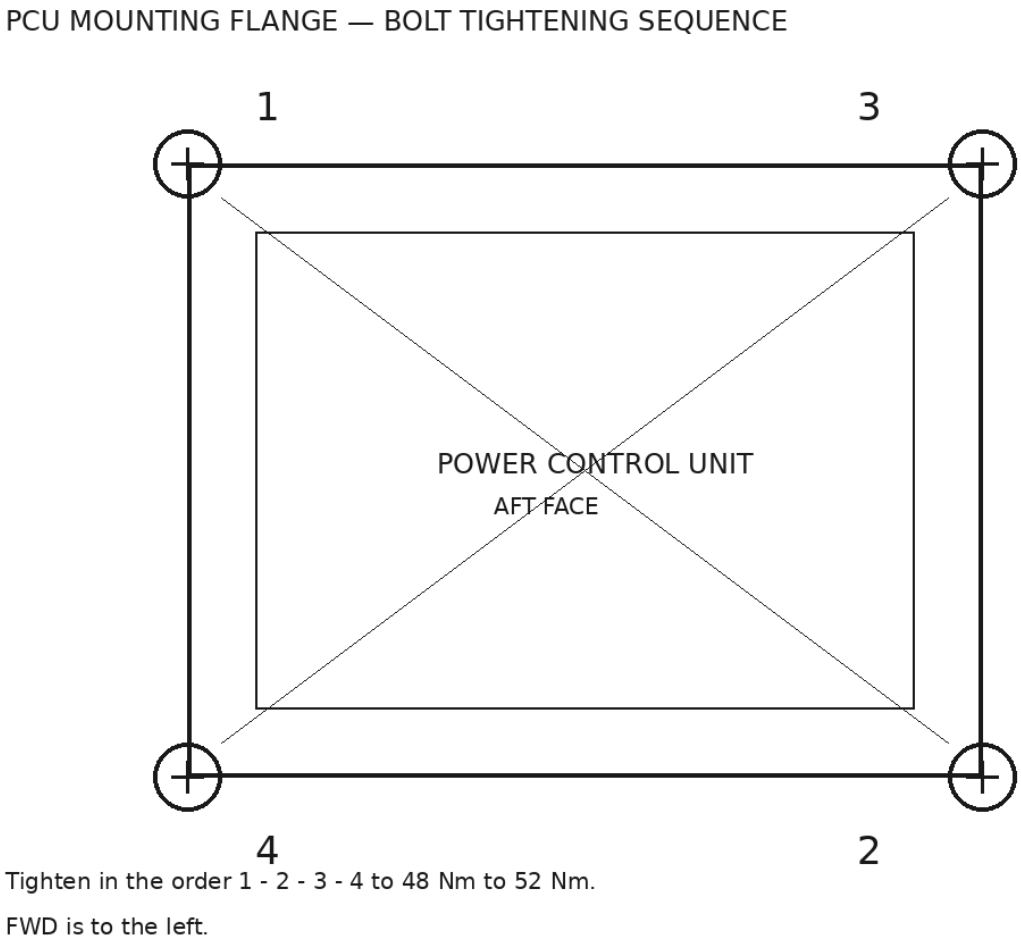
2. Install the power control unit

WARNING

Make sure that no person is below the unit while it is lifted into the belly fairing. The unit has a mass of 34 kg.

A. Lift the PCU into position and engage the two locating dowels on the support structure.

B. Install the four attachment bolts and the washers. Tighten the bolts to a torque of 48 Nm to 52 Nm in the sequence shown in Figure 1.



Power control unit — bolt tightening sequence — Figure 1

C. Install the lockwire on the four attachment bolts.

D. **APPLICABLE TO: MRDX-46 only**

Install the thermal blanket on the aft face of the unit and attach it with the six quick-release fasteners.

3. Connect the unit

A. Connect the green system supply and return lines to ports A and B. Tighten the couplings to 34 Nm to 38 Nm.

B. Connect the yellow system supply and return lines to ports C and D. Tighten the couplings to 34 Nm to 38 Nm.

C. Connect the electrical connectors 2027VC and 2028VC. Make sure that the coupling rings are fully engaged.

4. Do the installation test

A. Pressurize the green hydraulic system and examine the couplings on ports A and B for leakage. The permitted leakage is zero.

B. Do the slat system operational test and make sure that no fault message is shown on the maintenance terminal.

4. Close-up

Remove the handling sling and the platform from the work area.

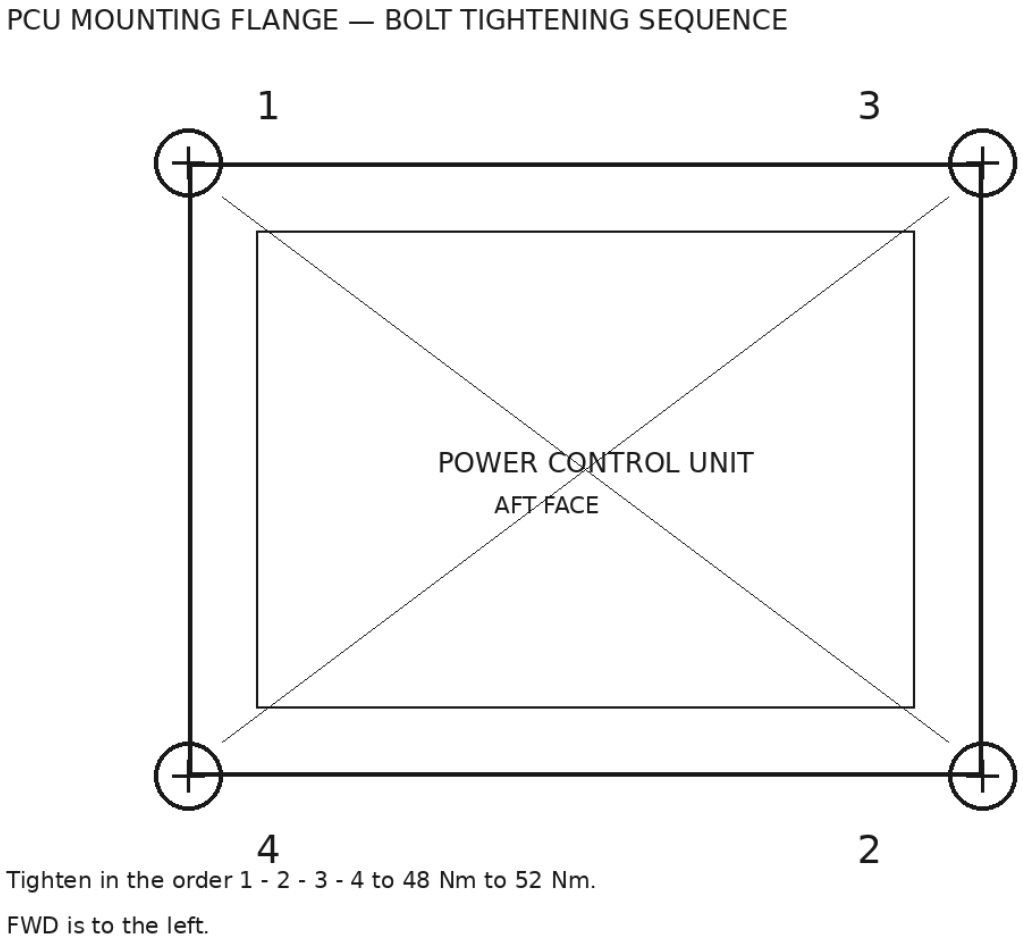
Close the fwd belly fairing access panel 196AB.

Make an entry in the aircraft technical log.

Slat actuation power control unit

Illustrated parts data

Object code	DMC-MRDX-A-27-81-00-00A-941A-A	**Issue**	004-00 (revised)
Issue date	2026-06-22	**Language**	en-GB
Responsible partner	AERALIS S.A.S. (U8025)	**Originator**	AERALIS S.A.S. — Spares Data Neuhafen (U8025)
Security	01 — Unclassified	**Quality assurance**	First verification (tabtop)
Applicability	MRDX-42 and MRDX-46, all aircraft		



Slat actuation power control unit — Figure 1

Parts list

FIG-ITEM	PART NUMBER	CAGE	NOMENCLATURE	UNITS	EFFECTIVITY
01-01000A	27DA1000000000	U8025	POWER CONTROL UNIT, SLAT ACTUATION	1	ALL

Parts list

FIG-ITEM	PART NUMBER	CAGE	NOMENCLATURE	UNITS	EFFECTIVITY
01-02000A	27DA1100000000	U8025	MOTOR, HYDRAULIC, VARIABLE DISPLACEMENT	2	ALL
01-03000A	27DA1110000000	U8025	VALVE BLOCK, SERVO CONTROL	1	ALL
01-04000A	NSA8102-118	81349	PACKING, PREFORMED	2	MRDX-42
01-04000B	NSA8102-119	81349	PACKING, PREFORMED, HIGH TEMPERATURE	2	MRDX-46
01-05000A	27DA1200000000	U8025	GEARBOX, DIFFERENTIAL	1	ALL
01-06000A	27DA1300000000	U8025	PICK-OFF UNIT, FEEDBACK POSITION Replaceable on the aircraft. See the removal data module.	2	ALL
01-07000A	NSA5075-08-032	81349	BOLT, HEXAGON HEAD, M8 x 32	4	ALL
01-08000A	NSA5305-08	81349	WASHER, PLAIN	4	ALL
01-09000A	27DA1400000000	U8025	BLANKET, THERMAL, AFT FACE	1	MRDX-46
01-10000A	27DB1000000000	U8025	SUPPORT STRUCTURE, PCU MOUNTING	REF	ALL